

where

$$K = \left\{ \sqrt{\frac{c_f}{2}} \left[5 Pr + 5 \ln(5 Pr + 1) - 14 + \sqrt{\frac{2}{c_f}} \right] \right\}^{-1}$$

Other analyses which assume a Stanton-number dependence on ϕ have been made in Refs. 17 and 21 and compared to experimental heat-transfer results for accelerated turbulent boundary-layer flows. In Ref. 17, the predictions exceeded the data by about 30% in part of the acceleration region, while in Ref. 21, the correspondence with the data was good.

The heat-transfer predictions shown in Fig. 15 as curve A are from Eq. (9b) for a stagnation temperature of 1500°R and a range of stagnation pressures from 45 to 254 psia, with the 18-in. cooled approach length. These predictions were made with properties evaluated as in Ref. 11 and conditions at the edge of the boundary layer determined from the wall static-pressure measurements. Shown as curve C in Fig. 15 is the prediction from Eq. (10), in which the friction coefficient $c_f/2$ was determined from the modified turbulent boundary-layer analysis. The reduction in the predicted heat-transfer coefficients provided by Eq. (9b) below the von Kármán analogy is due to the greater predicted thermal than velocity boundary-layer thicknesses through the nozzle. At the highest stagnation pressure, the predicted ratios of ϕ/θ as indicated in Fig. 16 are as large as 6 in the throat region. At the 75-psia stagnation pressure, the correspondence of the prediction from the modified turbulent boundary-layer analysis, Eq. (9b), with the data is good except near the nozzle exit. At the highest stagnation pressure of 254 psia, where the circumferential variation of the data is considerable, the correspondence with the averaged heat-transfer data is fair. The reproducibility of the data in Fig. 15 for 254 psia is indicated by the two sets of data shown by the open and shaded symbols. At the lowest stagnation pressure, $p_t = 44.8$ psia, the prediction exceeds the data by as much as 50% in the throat region. For the range of stagnation pressures, the predicted maximum value of the heat-transfer coefficient is just upstream of the throat, in agreement with the data.

The effect of temperature choice for property evaluation may be observed in Fig. 15 by comparing curves A and B. Curve B represents Eq. (9b) with properties evaluated at the film temperature T_f . In the throat region, it lies above the data but is in better agreement near the nozzle exit than curve A.

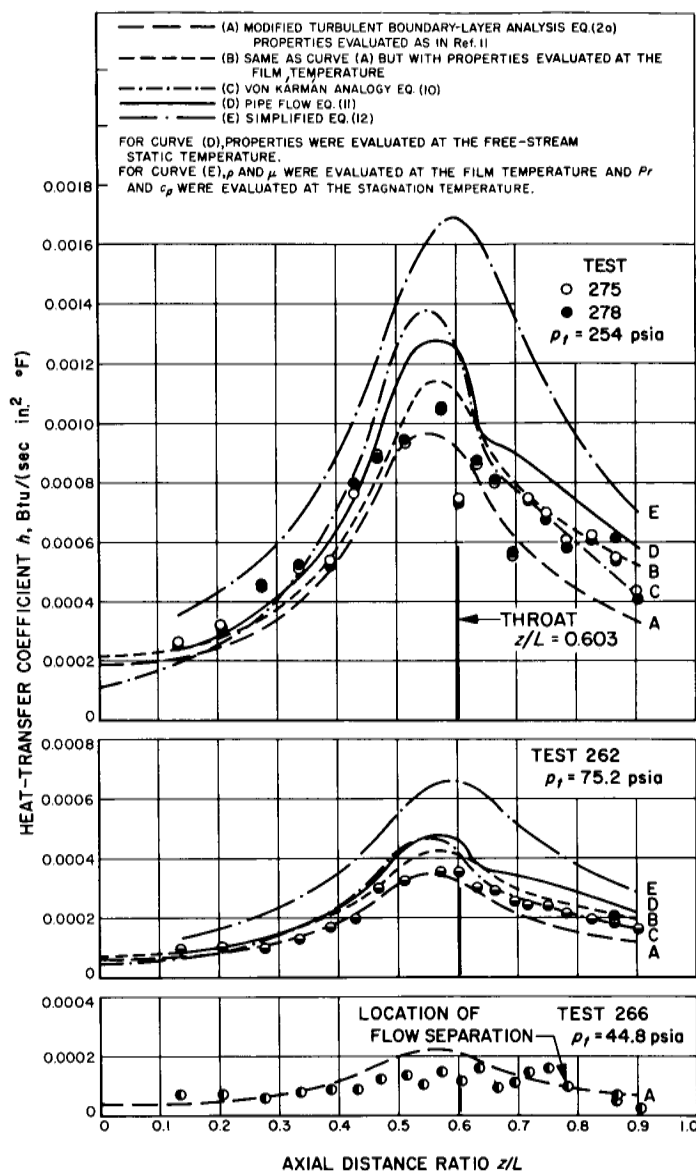


Fig. 15. Comparison of experimental heat-transfer coefficients with predictions at $T_{to} = 1500^\circ\text{R}$ with 18-in. cooled approach length

For comparison, the predictions from the following form of the pipe-flow equation for fully developed flow in which both the thermal and velocity boundary layer extend to the centerline and there is no significant pressure gradient are shown as curve D in Fig. 15.

$$St Pr^{0.6} = 0.023 Re_D^{-0.2} \quad (11)$$

Also shown, as curve E in Fig. 15, is the equation of Ref. 22: